

Table 1. Polygon-level spatial split statistics for AlphaEarth and Sentinel-2 datasets.

Partition	AlphaEarth	AlphaEarth	Sentinel-2	Sentinel-2
	Pixels	Polygons	Pixels	Polygons
Training	549,816 (74%)	4,387 (72%)	549,816 (74%)	4,387 (72%)
Validation	132,004 (18%)	1,097 (18%)	132,002 (18%)	1,097 (18%)
Test	64,856 (9%)	610 (10%)	64,856 (9%)	610 (10%)
Total	746,676	6,094	746,674	6,094

Table 2. Spectral indices computed from Sentinel-2 bands for each acquisition date.

Index	Formula	Biophysical Relevance	Reference
NDRE2 — Normalized Difference Red-Edge Index 2	$(B8 - B5) / (B8 + B5)$	Vegetation density	Frampton et al. (2013)
NDREI — Normalized Difference Red-Edge Index	$(B6 - B5) / (B6 + B5)$	Canopy chlorophyll content	Frampton et al. (2013)
RECI – Red-Edge Chlorophyll Index	$(B7 / B5) - 1$	Chlorophyll content	Frampton et al. (2013)
GCI - Green Chlorophyll Index	$(B7 / B3) - 1$	Green chlorophyll index	Frampton et al. (2013)
GNDVI - Green Normalized Difference Vegetation Index	$(B8 - B3) / (B8 + B3)$	Chlorophyll variation	Osgouei et al. (2019)
GI – Greenness Index	$B3 / B2$	Greenness index	Osgouei et al. (2019)
NDMI - Normalized Difference Moisture Index	$(B8 - B11) / (B8 + B11)$	Canopy moisture content	Frampton et al. (2013)
mNDWI - Modified Normalized Difference Water Index	$(B3 - B11) / (B3 + B11)$	Surface water detection	Osgouei et al. (2019)
NDTI - Normalized Difference Tillage Index	$(B11 - B12) / (B11 + B12)$	Tillage and bare soil	Osgouei et al. (2019)

Table 3. Hyperparameter search grids used in grid search cross-validation.

Classifier	Parameter	Values Searched
Random Forest	n_estimators	100, 200, 300
	max_depth	10, 20, None
	min_samples_leaf	1, 2
	max_features	sqrt, log2
Decision Tree	max_depth	10, 20, 30
	min_samples_split	2, 5
	min_samples_leaf	1, 2
XGBoost	n_estimators	100, 200
	learning_rate	0.05, 0.1
	max_depth	4, 6
	Subsample	0.8, 1.0
	colsample_bytree	0.6, 0.8
LightGBM	n_estimators	100, 200
	learning_rate	0.05, 0.1
	num_leaves	31, 63
	max_depth	-1, 10
LinearSVC	C	0.1, 1, 10

Table 4. Accuracy metrics of the implemented experiments.

Dataset	ID	Model	Overall Accuracy	Balanced Accuracy	Precision (W)	F1 Weighted	F1 Macro	Cohen Kappa	MCC
AlphaEarth	1	LGBMClassifier	0.9579	0.8282	0.9581	0.9566	0.8510	0.9272	0.9274
	2	XGBClassifier	0.9556	0.8088	0.9557	0.9541	0.8381	0.9232	0.9234
	3	RandomForestClassifier	0.9530	0.8045	0.9535	0.9508	0.8354	0.9186	0.9188
	4	LinearSVC	0.9437	0.8130	0.9446	0.9424	0.8142	0.9027	0.9029
	5	DecisionTreeClassifier	0.9209	0.7522	0.9210	0.9205	0.7439	0.8641	0.8641
Sentinel 2 Indices	6	XGBClassifier	0.9149	0.7545	0.9119	0.9118	0.7700	0.8530	0.8532
	7	RandomForestClassifier	0.9134	0.7482	0.9106	0.9107	0.7641	0.8506	0.8508
	8	LGBMClassifier	0.8768	0.7930	0.9055	0.8864	0.7166	0.7977	0.8005
	9	DecisionTreeClassifier	0.8332	0.7294	0.8780	0.8500	0.6436	0.7322	0.7372
	10	LinearSVC	0.8149	0.7143	0.8675	0.8322	0.6254	0.7039	0.7100
Sentinel 2 Bands	11	XGBClassifier	0.9295	0.7646	0.9260	0.9260	0.7936	0.8775	0.8778
	12	LGBMClassifier	0.9236	0.8092	0.9245	0.9229	0.7797	0.8687	0.8689
	13	RandomForestClassifier	0.9243	0.7690	0.9207	0.9212	0.7840	0.8686	0.8689
	14	DecisionTreeClassifier	0.8493	0.7324	0.8881	0.8654	0.6587	0.7514	0.7537
	15	LinearSVC	0.8438	0.6950	0.8824	0.8580	0.6169	0.7441	0.7471
Sentinel 2 Bands+Indices	16	LGBMClassifier	0.9329	0.8118	0.9325	0.9316	0.8114	0.8841	0.8842
	17	RandomForestClassifier	0.9310	0.7894	0.9290	0.9286	0.7903	0.8805	0.8807
	18	XGBClassifier	0.9316	0.7758	0.9284	0.9282	0.7997	0.8812	0.8815
	19	LinearSVC	0.8973	0.7710	0.9104	0.8996	0.7098	0.8260	0.8266
	20	DecisionTreeClassifier	0.8961	0.7399	0.9036	0.8992	0.7000	0.8230	0.8232

Table 5. Class-wise accuracy metrics of top 6 performing experiments.

Class	ID 1			ID 2			ID 3			ID 4			ID 16			ID 17		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Hazelnut	0.8551	0.9214	0.8871	0.8610	0.9214	0.8902	0.8475	0.9428	0.8926	0.7597	0.8980	0.8231	0.8679	0.8715	0.8697	0.8420	0.8652	0.8535
Forest	0.9891	0.9977	0.9934	0.9890	0.9982	0.9936	0.9875	0.9988	0.9931	0.9883	0.9954	0.9919	0.9865	0.9935	0.9900	0.9842	0.9933	0.9888
Permanent crops	0.8223	0.8031	0.8126	0.8499	0.7877	0.8176	0.8588	0.7928	0.8245	0.6709	0.7687	0.7165	0.6757	0.7291	0.7014	0.6857	0.6406	0.6624
Grassland	0.7611	0.8240	0.7913	0.7479	0.7940	0.7702	0.7282	0.7790	0.7528	0.6911	0.7535	0.7210	0.5940	0.6947	0.6404	0.6412	0.6625	0.6517
Sparsely vegetated areas	0.8963	0.5803	0.7045	0.8796	0.5728	0.6938	0.9115	0.4912	0.6384	0.8092	0.4737	0.5975	0.6780	0.4241	0.5218	0.7299	0.3934	0.5112
Arable land	0.8990	0.8728	0.8857	0.8784	0.8750	0.8767	0.8629	0.8568	0.8598	0.8871	0.8116	0.8477	0.8211	0.7498	0.7838	0.7775	0.7899	0.7837
Urban fabric	0.9908	0.9887	0.9898	0.9862	0.9899	0.9881	0.9845	0.9959	0.9901	0.9856	0.9814	0.9835	0.9573	0.9664	0.9618	0.9441	0.9732	0.9584
Road and rail networks	0.9427	0.9583	0.9504	0.9577	0.9456	0.9516	0.9670	0.9347	0.9506	0.9272	0.9368	0.9320	0.8772	0.9297	0.9027	0.9187	0.9121	0.9154
Water courses	0.9696	0.6566	0.7830	0.9868	0.5918	0.7399	0.9946	0.5870	0.7383	0.9221	0.7120	0.8036	0.9932	0.9288	0.9599	0.9730	0.8544	0.9099
Water bodies	0.9022	0.9950	0.9464	0.8832	0.9955	0.9360	0.8836	0.9959	0.9364	0.9244	0.9874	0.9549	0.9786	0.9923	0.9854	0.9563	0.9860	0.9709
Wetlands	0.7736	0.5125	0.6165	0.8293	0.4250	0.5620	0.8636	0.4750	0.6129	0.5495	0.6250	0.5848	0.5714	0.6500	0.6082	0.4050	0.6125	0.4876
Accuracy	0.9579	0.9579	0.9579	0.9556	0.9556	0.9556	0.9530	0.9530	0.9530	0.9437	0.9437	0.9437	0.9329	0.9329	0.9329	0.9310	0.9310	0.9310
Macro avg	0.8911	0.8282	0.8510	0.8954	0.8088	0.8381	0.8991	0.8045	0.8354	0.8286	0.8130	0.8142	0.8183	0.8118	0.8114	0.8052	0.7894	0.7903
Weighted avg	0.9581	0.9579	0.9566	0.9557	0.9556	0.9541	0.9535	0.9530	0.9508	0.9446	0.9437	0.9424	0.9325	0.9329	0.9316	0.9290	0.9310	0.9286